

Kalman-Filtered Trend Trader: A Simple Portfolio Strategy

final report

1 Project Idea

Financial markets show a lot of daily movement, but not all of it is meaningful. Many price changes are simply noise. The goal of this project was to build a system that focuses on the actual direction of the market instead of reacting to every small fluctuation.

The portfolio includes Bitcoin – High volatility, high risk, Nifty 50 – Indian equity market benchmark, Gold – Hedge / defensive asset, Cash – Risk-free holding (capital not allocated elsewhere). Each asset represents a different level of risk. Cash is treated as an active choice, since choosing not to invest is also a valid decision.

2 Strategy Logic

meaningful decisions should be based on clear direction rather than short-term market noise. Financial markets move every day, but most of these movements do not represent real trends. The system therefore focuses on identifying steady and persistent movement before taking action.

Each asset in the portfolio is first analyzed independently. Instead of comparing raw prices, the strategy attempts to understand whether an asset is generally moving upward, downward, or behaving randomly. This prevents short-term fluctuations from dominating the decision-making process.

Sudden price jumps or drops are treated with caution. The strategy does not react immediately to sharp movements unless they continue consistently over time. This helps avoid panic-driven decisions and unnecessary trading.

Once the system has a sense of how confident it is in the movement of each asset, capital is allocated accordingly. Assets showing strong and stable upward movement receive higher allocation, while assets with weak or unclear behavior receive less. If none of the assets appear reliable, the strategy allocates capital to cash. Holding cash is treated as an active and meaningful decision rather than a default state.

Allocation decisions are made at the portfolio level rather than for individual assets in isolation. The strategy considers where capital is best placed relative to all available options, allowing it to balance growth and safety dynamically.

To ensure realistic behavior, transaction costs are applied whenever allocations change. This discourages frequent rebalancing and encourages the system to hold positions until there is a clear reason to change them.

Finally, all decisions are executed with a one-day delay. The strategy only uses information available at the current time and applies decisions in the future, ensuring causality and preventing unrealistic assumptions.

Overall, the strategy behaves like a patient and disciplined investor, acting only when there is sufficient confidence and remaining cautious during uncertain periods.

3 Role of Kalman Filtering

Raw market prices are noisy and can lead to poor decisions if used directly. Kalman filtering helps by removing random fluctuations and estimating the true direction and strength of movement.

The strategy does not rely on raw prices. Instead, it uses the filtered trend signal to decide how to allocate capital. This results in smoother behavior and fewer unnecessary trades.

4 Evaluation and Results

The strategy was tested using a causal backtesting setup. Decisions at a given time were based only on information available at that time and were applied the next day. A transaction cost of 0.1% was included.

Performance was measured using total returns, risk, risk-adjusted performance, and maximum losses. The strategy was compared against a Nifty 50 buy-and-hold benchmark.

Overall, the portfolio showed smoother growth, reduced drawdowns during volatile periods, and better control over trading frequency.

5 Future Improvements

Possible improvements include using a learning-based agent for allocation decisions, adjusting allocations based on market risk, and testing the strategy across different market conditions. More advanced models could also be explored to better capture complex market behavior.

6 Conclusion

This project demonstrates that removing noise before making decisions leads to calmer and more reliable trading strategies. Intelligent systems do not react to every movement but act only when there is a clear and consistent signal.